COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 - Data Structure Algorithms	**SECTION:**	CpE21S2
DATE PERFORMED:	9/3/2026	**DATE SUBMITTED:**	9/3/2026
NAME:	Demain, Jaime Luis Esguerra, Paula Esteta, Jesse Matthew Estorco, Clint Hoven Fernandez, Cedric	**INSTRUCTOR:**	Engr. Jimlord Quejado

SEATWORK 7.1 Using sorting algorithms

1. PROCEDURE OUTPUT

Answer the following questions:

1. **What is a sorting algorithm?**
    - A sorting algorithm is any algorithm that sorts data based on a certain standard method of comparison. Frequently, the use of deterministic algorithms can greatly improve the efficiency of other algorithms such as searching, pattern matching and data analysis. Sorting paradigms are categorized based on the operational parameters such as sorting comparison or non-sorting and other properties including algorithmic stability and memory consumption. An asymptotic analysis determining the optimal comparison-based sorting performed by the operation achieving an average and worst-case complexity of O(n \log n) is used to evaluate the theoretical performance. As such, the algorithms are at the heart of computer science and power a wide range of basic database indexing operations, basic system software and many complex data pipeline architectures.
2. **Where can sorting algorithms be used?**
    - Sorting algorithms are most when it is required to organize information to make it easier to find and display data for example browsers and search engines utilize this to arrange the search by

relevance, online shops use it to organize product from highest price to lowest making this algorithm essential for speeding up searches and improving not only the user but also the programmers convenience.

3. **Explain the different types of sorting algorithms.**

Selection Sort	It sorts by repeatedly selecting the smallest or the largest element from a particular portion and swapping the element with the first unsorted element.
Bubble Sort	Repeatedly swapping the adjacent element if they are in the wrong order.
Insertion Sort	This works in iteratively inserting each element of an unsorted list into its correct position in a sorted portion of the list.
Merge Sort	By recursively dividing the input array into two halves, recursively sorting the two halves and finally merging them back together to obtain the sorted array.
Quick Sort	An element is seen as a pivot where it partitions the given array around the picked pivot by placing the pivot in its correct position in the sorted array.
Heap Sort	It finds the maximum or minimum element and swaps the element with the last or first element.
Cycle Sort	Divide the array into cycles then each cycles consists of elements that belong to each other's correct positions and rotate these cycles to place all elements correctly.
3-way Merge Sort	To extend merge sort by dividing the array into three parts instead of two. First, recursively sort the three subarrays. Then, merge them by selecting the smallest element among the three at each step. Temporary arrays store the divided parts, and a comparison process finds the minimum to insert into the original array.
Counting Sort	To count the frequency of each distinct element in the input array and use that information to place the elements in their correct sorted positions.
Radix Sort	By repeatedly sorting the elements by their significant digits, from the least significant to the most significant, it achieves the final sorted order.
Bucket Sort	Bucket sort is a sorting technique that involves dividing elements into various groups, or buckets. These buckets are formed by uniformly distributing the elements. Once the elements are divided into buckets, they can be sorted using any other sorting algorithm. Finally, the sorted elements are gathered together in an ordered fashion.

4. **Give sample programs in C++ that uses sorting algorithms, specifically selection sort, insertion sort, and bubble sort. Explain how the programs work.**

- **Selection Sort - Student Ranking System**

Selection Sort can be used in a student ranking program to arrange students based on their grades (sorting is highest to lowest) . It repeatedly finds the highest grade from the unsorted list and swap places with the compared value until all students are ranked.

- **Insertion Sort - Contact List**

Insertion Sort can be used in a contact list program to arrange names alphabetically. When a new contact is added, the algorithm compares the name with the existing names on the list and inserts it into the correct position.

- **Bubble Sort - Online Shopping System**

Bubble Sort can be used in an online shopping program to arrange products according to their prices (sorting is cheapest to most expensive). It compares two prices that is next to each other and swaps them if the other is cheaper than the one on the left until the products are arranged from the lowest to highest price.

COMPUTER ENGINEERING

QUEZON CITY

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

Lab activities, templates, and related instructional materials shall not be reuploaded, reproduced, distributed, or shared with external organizations or third parties without the prior express written permission of the Technological Institute of the Philippines.